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1. Deploying PostgreSQL in a linux Container

Objective: Spin up a systemd-enabled CentOS container, install PostgreSQL 14 from the official Red Hat repositories, and initialize a new database with specific UTF-8 encodings.

Phase 1: Provision the Container

Because PostgreSQL relies on `systemd` (`systemctl`) to manage its background services, the container must be launched with elevated privileges and the `/usr/sbin/init` command.

Execute this from your host machine terminal:

```
#docker run --name postgresdb0 --hostname postgresdb0 -p 5432:5432 -p 5411:5411 -i -t --privileged --network spark-net  
-v /Users/Xxx/Documents/Software:/Software arm64v8/oraclelinux:9 /usr/sbin/init
```

<https://www.postgresql.org/download/linux/redhat/>

Once the container is running, open a new terminal and execute a bash shell inside it:

```
#dnf install langpacks-en glibc-all-langpacks -y
```

Install PostgreSQL

Execute the following commands inside your CentOS container as the `root` user.

```
# dnf install postgresql-server
# postgresql-setup --initdb
```

Phase 3: Initialize and Start the Service

Before PostgreSQL can start, it needs an initialized database cluster (the physical files on disk).

Step 1: Initialize the Database

```
systemctl enable postgresql
systemctl start postgresql
systemctl status postgresql
```

Log in to Postgres and enter the command '\l+' to view the list of databases in PostgreSQL terminal `psql`:

```
dnf install sudo
sudo -u postgres psql
ALTER USER postgres PASSWORD 'postgres';
```

```
[root@rheltest ~]# su - postgres
[postgres@rheltest ~]$ psql
psql (12.5)
Type "help" for help.
postgres=# \l+
```

Create a Database

Create a new database with strict UTF-8 encoding:

CREATE DATABASE myDatabase WITH ENCODING 'UTF8' ;

Connect to the new database using the command:

\c mydatabase

<https://www.redhat.com/sysadmin/postgresql-setup-use-cases>

Change root password

#passwd

Enter root123

Allow all to connect to postgresDB from remote server by updating the following contents:

*#vi /var/lib/pgsql/data/**postgresql.conf***

listen_addresses = ''*

Append in the last line of the following file if doesn't exist or update to have the same entries.

```
# vi /var/lib/pgsql/data/pg_hba.conf
```

```
local  all         all         trust
host   all         all         127.0.0.1/32  trust
host   all         all         all           trust
```

Restart the postgresdb server

```
systemctl restart postgresql
```

On PSQL

```
#SHOW work_mem;
```